

Lex: A Programming Language for Developing Enforceable Legal Formalizations

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1 Introduction

2 Enforcement under refinement

Definition 1. Let $\Phi = (\Sigma, \kappa, \phi)$ and $\Phi' = (\Sigma', \kappa', \phi')$ be two specifications. We say that Φ refines Φ' under assumption ψ iff $\exists \pi : \mathbb{E}(\Sigma) \rightarrow \text{MFOTL}(\Sigma')$ such that

- (i) $v, i \models_{\pi(\sigma)} \phi' \wedge \psi \implies v, i \models_{\sigma} \phi$
- (ii) $\kappa' \vdash \pi(e) : \kappa(e)$

Theorem 1. If Φ

3 Logical foundations

4 Syntax

4.1 Modules

Every Lex file defines a module, while folders containing Lex files define groups of modules. The syntax of module imports is as follows:

$$\begin{aligned} \text{import} &::= \text{import } [\text{formex} \mid \text{akomaNtoso}] \text{idents} \\ \text{idents} &::= \text{ident} \mid \text{idents}.\text{ident} \end{aligned}$$

The special flags **formex** and **akomaNtoso** instruct the Lex compiler to load the corresponding modules as XML files containing a machine-readable version of the law, rather than Lex files. Two formats are supported: FORMEX (the standard EU format for machine-readable legal documents) and Akoma Ntoso (an international standard used, e.g., in Switzerland).

When the target module is a Lex module, the sequence $\text{ident}_1 \dots \text{ident}_n$ refers to the file $\text{ident}_1 / \dots / \text{ident}_n.\text{lex}$; if the target module is an XML module in FORMEX or Akoma Ntoso format, the file is $\text{ident}_1 / \dots / \text{ident}_n.\text{xml}$.

4.2 Labels

Labels are used to capture the structure of the law. Their syntax is as follows:

$$\begin{aligned} \text{label_level} &::= \text{law} \mid \text{title} \mid \text{chapter} \mid \text{section} \\ &\quad \mid \text{article} \mid \text{paragraph} \mid \text{point} \mid \text{subpoint} \\ \text{label} &::= \text{label_level}[\text{int}] \text{ string } [\text{string}] \end{aligned}$$

The optional integer parameter that follows the level can be used to encode an arbitrary number of additional hierarchical levels: `chapter[1]` is one level below `chapter` but still above `section`, `chapter[i]` is one level below `chapter[i - 1]` and one level above `chapter[i + 1]`, etc. The second and (optional) third argument collect the number and header of the corresponding part of the law.

Example 1. Art. 2 GDPR can be introduced as

$$\text{article "2" "Material scope"}$$

The levels `law`, `article`, `paragraph`, `point`, and `subpoint` plain a special role in determining the *qualified name* of `Lex` rules. Rules can only defined within articles, which can themselves only defined within laws. The induced uniqueness constraints are shown in Table 1. The qualified name of a rule is of the form

$$\text{law article}[(\text{paragraph})][(\text{point})][(\text{subpoint})] \# \text{rule_id}$$

where additional level numbers placed in brackets can be introduced if intermediate, indexed levels are present. Rules introduced within the scope of a given label can be referred to using a suffix of their qualified name that omits the part determined by the scope. The set of all rules located within the scope of a given label can be referred to using the prefix of their qualified name corresponding to that scope, or by a suffix of this prefix, if applicable within the current scope.

Example 2. The set of rules defined in Art. 2, §2 of the GDPR can be referred to as `"GDPR 2(2)"`. Within the GDPR, they can be referred to as just `"2(2)"`, and within Art. 2, as `"(2)"`. The first of such rules is `"GDPR 2(2) #1"`, the second one is `"GDPR 2(2) #2"`, etc. The rule IDs `1..k` can be overwritten by defining explicit rule IDs (see the syntax of rules below).

4.3 Types

`Lex` has four base types: `int` (integers), `float` (floating-point numbers), `bool` (booleans), and `str` (strings). New types can be declared as follows:

$$\text{type_decl} ::= \text{type ident } [[\text{is} \mid \text{subtype}] \text{ ident}]$$

When another, already existing type name is specified after the `is` keyword, the newly declared type is created as a copy of that existing type. As a consequence,

Label	level	Number unique within	Part of qualified name	Required
law	—	✓		✓
title	law			
chapter	law			
section	law			
article	law	✓		✓
paragraph	article	✓		
point	paragraph	✓		
subpoint	point	✓		

Table 1: Uniqueness constraints for labels

the new type will have the same internal representation and take the same values as the existing type. However, the existing type and the new type will still be considered distinct types by Lex’s type system (see example below). In contrast, the `subtype` keyword support specifying a new type as a subtype of an existing type. In this case, the new type will have both the same internal representation and the capacity to be used in any context where the existing type could be used.

Example 3. Consider the following declarations:

```
type id is string
type user_id subtype id
```

Further, consider an event with signature $A(b:id, c:string)$ and a variable `uid` of type `user_id`. The expression $A(uid, "foo")$ is type-correct, since `user_id` is a subtype of `id`. The expression $A(uid, uid)$ is not type-correct, since `user_id` is not a subtype of `string` (because `id` is not a subtype of `string`). On the other hand, $A("foo", "foo")$ is type-correct since the type `id` can contain string values.

4.4 Events

The syntax for event declarations is as follows:

```
capabilities ::= observable | suppressable | causable | internal
               | observable causable | causable observable
               | suppressable causable | causable suppressable
event_decl ::= capabilities event ident
               [""event_desc""]
               ident11 : ident12
               ...
               identk1 : identk2
```

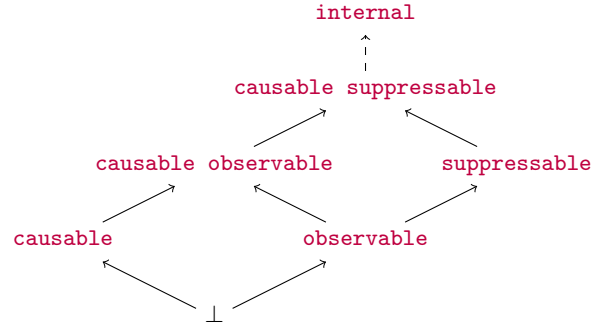


Fig. 1: Capability lattice

where *event_desc* is a string that can contain special placeholders $\{ident_{i1}\}$ for $1 \leq i \leq k$. The capabilities associated to an event are organized into the following lattice, where $a \rightarrow b$ denotes that *a* is strictly weaker than *b*.

The pairs of identifiers $ident_{i1} : ident_{i2}$ contained in the event declaration determine an (ordered) list of event parameters ($ident_{i1}$) and their types ($ident_{i2}$). Finally, the event description *event_desc* provides the high-level semantics of the event, which should refer to the parameters $ident_{i1}, \dots, ident_{k1}$ using the $\{ident_{i1}\}$ syntax.

Example 4. The following code declares a new suppressable event capturing personal data processing in a system, assuming three previously defined types `processing_id`, `processor_id`, and `data_id`.

```

suppressable event PersonalDataProcessing
"""
    processor {x} is processing personal data {z} as part
    of processing operation {ep}
"""
    ep : processing_id
    x  : processor_id
    z  : data_id

```

Event capabilities There are two classes of event capabilities: **internal** and all other capabilities. Choosing a capability (other than internal) depends on two factors: 1. whether the event can actually be observed, caused, suppressed, etc. and 2. what the enforcer requires as a "minimal" capacity in order for the rules where the event is used to be enforced.

observable

- * The event can be observed by the enforcer.
- * e.g. an event might not be observable if it is not always logged when it happens (this is highly dependent on the system in question).

causable

- * The event can be caused by the enforcer.

suppressable

- * The event can be suppressed by the enforcer. (every occurrence of the event is logged)
- * **suppressable** events are **observable** by definition. (an event cannot not be suppressed if it was never observed)

observable causable or **causable observable**

- * combines the capabilities of **observable** and **causable**

suppressable causable or **causable suppressable**

- * The event can be suppressed and caused by the enforcer.
- * This capability inherits observability from **suppressable**

internal events serve a special purpose in the Lex language.

- * They occur on the right-hand side of constitutive rules (after the **constitute** keyword).
- * They are not "emitted" by the system, rather they can be seen as placeholders for a combination of other events, as defined by constitutive rules.
- * Completely hidden from the programmer, they are also used to internally create special events for scope and exception rules.

4.5 Rules

The general syntax for rule declarations is as follows:

```

    rule_verb_action ::= oblige
    rule_verb_constitute ::= constitute
    rule_verb_relative ::= except | scope
    rule_constr_verb ::= causing | suppressing
    rule_constr ::= rule_constr_verb ident1 ... identm
    rule_action ::= rule_verb_action [patt2]
                    [ident41 :] expr41
                    ...
                    [ident4k4 :] expr4k4
    rule_reference ::= { label_level1[ [int1] ] string1 ... label_levelk[ [intk] ] stringk [rule stringk+1] }
    rule_relative ::= rule_verb_relative [patt2]
                    rule_reference1
                    ...
                    rule_referencel
    rule_decl ::= rule [string]
                [fix
                 ident11 : ident21
                 ...
                 ident1n : ident2n]
                [whenever [patt1]
                 [ident31 :] expr31
                 ...
                 [ident3k3 :] expr3k3]
                (rule_action | rule_relative)
                [[transparently] enforceable] rule_constr1, ..., rule_constrl

```

In the following we review the syntax of expressions $expr_{ij}$ (Section 4.5), the available verbs (Section 4.5) and the temporal patterns determined by $patt_1$ and $patt_2$ (Section 4.5).

Expressions. The syntax of expressions closely follows MFOTL syntax:

$$\begin{aligned}
 \text{expr_itv} &::= [\text{time_bound}_1, (\text{time_bound}_2 \mid *)] \\
 \text{term} &::= (\text{const} \mid \text{ident})[:\text{ident}] \\
 \text{pred} &::= \text{ident}(\text{term}_1, \dots, \text{term}_k) \\
 \text{expr} &::= \text{TRUE} \mid \text{FALSE} \mid \text{pred} \mid \text{special_pred} \\
 &\quad \mid \text{NOT } \text{expr} \mid \text{expr}_1 (\text{AND} \mid \text{OR} \mid \text{IMPLIES} \mid \text{EQUIV}) \text{expr}_2 \\
 &\quad \mid (\text{EXISTS} \mid \text{FORALL}) \text{expr} \\
 &\quad \mid (\text{ONCE} \mid \text{EVENTUALLY} \mid \text{HISTORICALLY} \mid \text{ALWAYS}) [\text{expr_itv}] \text{expr} \\
 &\quad \mid \text{expr}_1 (\text{SINCE} \mid \text{UNTIL} \mid \text{TRIGGER} \mid \text{RELEASE}) [\text{expr_itv}] \text{expr}_2 \\
 \text{special_pred} &::= \text{obliged}(\text{pred}) \mid \text{permitted}(\text{pred}) \mid \text{prohibited}(\text{pred}) \\
 &\quad \mid \text{applicable}(\text{string}) \mid \text{satisfied}(\text{string}) \mid \text{violated}(\text{string})
 \end{aligned}$$

Verbs.

Temporal patterns. Except for **except**, **scope**, and **constitute** rules, every Lex rule declaration represents a statement of the form

$$\begin{aligned}
 &\text{FORALL } \text{ident}_{11}, \dots, \text{ident}_{1n}. \\
 &\quad \text{ALWAYS}(\text{form}_1(\text{expr}_{11} \text{ AND } \dots \text{ AND } \text{expr}_{1k}) \\
 &\quad \text{IMPLIES } \text{form}_2(\text{expr}_{21} \text{ AND } \dots \text{ AND } \text{expr}_{2k}))
 \end{aligned}$$

where form_1 and form_2 are determined by patt_1 and patt_2 , respectively. These options use the following grammars for time and time intervals:

$$\begin{aligned}
 \text{time_unit} &::= \text{y} \mid \text{d} \mid \text{h} \mid \text{s} \\
 \text{time_bound} &::= \text{int } \text{time_unit} \\
 \text{time_interval} &::= \text{within } \text{time_bound} \mid \text{before } \text{time_bound} \mid \text{after } \text{time_bound}
 \end{aligned}$$

Table 2 lists available patt_i and the corresponding form_i functions, as well as the conversion of time_interval expressions into intervals:

Typing Rules of Temporal Patterns

5 Compilation

5.1 To MFOTL

5.2 To LegalRuleML

5.3 To documentation

References

$$\begin{array}{c}
\frac{\Gamma \vdash \phi_0 : \mathbb{C} \quad \dots \quad \Gamma \vdash \phi_n : \mathbb{C}}{\Gamma \vdash \bigwedge_{i=0}^n \phi_i : \mathbb{C}} \text{Present}^{\mathbb{C}} \quad \frac{\Gamma \vdash \phi_j : \mathbb{S}}{\Gamma \vdash \bigwedge_{i=0}^n \phi_i : \mathbb{S}} \text{Present}^{\mathbb{S}_j} \\
\\
\frac{\Gamma \vdash \phi_0 : \mathbb{C} \quad \dots \quad \Gamma \vdash \phi_n : \mathbb{C}}{\Gamma \vdash \text{EVENTUALLY}_{[a,b]} \bigwedge_{i=0}^n \phi_i : \mathbb{C}} \text{Eventually}^{\mathbb{C}} \quad \frac{\Gamma \vdash \phi_j : \mathbb{S}}{\Gamma \vdash \text{EVENTUALLY}_{[a,b]} \bigwedge_{i=0}^n \phi_i : \mathbb{S}} \text{Eventually}^{\mathbb{S}_j} \\
\\
\frac{\Gamma \vdash \phi_0 : \mathbb{C} \quad \dots \quad \Gamma \vdash \phi_n : \mathbb{C}}{\Gamma \vdash \text{ALWAYS}_{[a,b]} \bigwedge_{i=0}^n \phi_i : \mathbb{C}} \text{Always}^{\mathbb{C}} \quad \frac{\Gamma \vdash \phi_j : \mathbb{S}}{\Gamma \vdash \text{ALWAYS}_{[a,b]} \bigwedge_{i=0}^n \phi_i : \mathbb{S}} \text{Always}^{\mathbb{S}_j} \\
\\
\frac{b \neq \infty \quad \Gamma \vdash \phi_0 : \mathbb{C} \quad \dots \quad \Gamma \vdash \phi_n : \mathbb{C}}{\Gamma \vdash \psi \text{Until}_{[0,b]} \bigwedge_{i=0}^n \phi_i : \mathbb{C}} \text{Until}^{\text{CR}} \quad \frac{\Gamma \vdash \phi_j : \mathbb{S}}{\Gamma \vdash \psi \text{Until}_{[a,b]} \bigwedge_{i=0}^n \phi_i : \mathbb{S}} \text{Until}^{\mathbb{S}_j} \\
\\
\frac{b \neq \infty \quad \Gamma \vdash \psi : \mathbb{C} \quad \Gamma \vdash \phi_0 : \mathbb{C} \quad \dots \quad \Gamma \vdash \phi_n : \mathbb{C}}{\Gamma \vdash \psi \text{Until}_{[a,b]} \bigwedge_{i=0}^n \phi_i : \mathbb{C}} \text{Until}^{\text{CLR}} \\
\\
\frac{\Gamma \vdash \phi_j : \mathbb{S}}{\Gamma \vdash \text{HISTORICALLY}_{[a,b]} \bigwedge_{i=0}^n \phi_i : \mathbb{S}} \text{Historically}^{\mathbb{S}_j} \\
\\
\frac{0 \in [a,b] \quad \Gamma \vdash \phi_0 : \mathbb{C} \quad \dots \quad \Gamma \vdash \phi_n : \mathbb{C}}{\Gamma \vdash \text{Once}_{[a,b]} \bigwedge_{i=0}^n \phi_i : \mathbb{C}} \text{Once}^{\mathbb{C}} \quad \frac{0 \in [a,b] \quad \Gamma \vdash \psi : \mathbb{C}}{\Gamma \vdash \left(\bigwedge_{i=0}^n \phi_i \right) \text{Since}_{[a,b]} \psi : \mathbb{C}} \text{Since}^{\mathbb{C}} \\
\\
\frac{0 \notin [a,b] \quad \Gamma \vdash \phi_j : \mathbb{S}}{\Gamma \vdash \left(\bigwedge_{i=0}^n \phi_i \right) \text{Since}_{[a,b]} \psi : \mathbb{S}} \text{Since}^{\text{SL}_j} \quad \frac{0 \in [a,b] \quad \Gamma \vdash \phi_j : \mathbb{S} \quad \Gamma \vdash \psi : \mathbb{S}}{\Gamma \vdash \left(\bigwedge_{i=0}^n \phi_i \right) \text{Since}_{[a,b]} \psi : \mathbb{S}} \text{Since}^{\text{SLR}_j}
\end{array}$$

Fig. 2: Typing rules for temporal patterns

<i>itv</i>	<i>I</i>	Restriction
–	$[0, \infty)$	
within <i>b</i>	$[0, b]$	
before <i>a</i>	$[a, \infty)$	Only past patterns
strictly before <i>a</i>	(a, ∞)	Only past patterns
after <i>a</i>	$[a, \infty)$	Only future patterns
strictly after <i>a</i>	(a, ∞)	Only future patterns
between <i>a</i> and <i>b</i>	$[a, b]$	
strictly between <i>a</i> and <i>b</i>	(a, b)	
between <i>a</i> and <i>b</i> excluded	$[a, b)$	
between <i>a</i> excluded and <i>b</i>	$(a, b]$	

Description <i>patt_i</i>	<i>form_i</i> (ϕ)	Type
–	ϕ	Present
eventually [<i>itv</i>]	EVENTUALLY _{<i>I</i>} ϕ	Future
eventually delaying if ψ [<i>itv</i>]	ψ UNTIL _{<i>I</i>} ϕ	Future
always in the future [<i>itv</i>]	ALWAYS _{<i>I</i>} ψ	Future
once [<i>itv</i>]	ONCE _{<i>I</i>} ψ	Past
always in the past [<i>itv</i>]	HISTORICALLY _{<i>I</i>} ψ	Past
always since ψ [<i>itv</i>]	ϕ SINCE _{<i>I</i>} ψ	Past

Table 2: Available temporal patterns for Lex rules